

CDGL–YOLOv11Lite: A Lightweight Field Maize–Seedling Weed Detector Based on Cross-Domain Shared Attention and a Conditional Gated Linear Unit

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Abstract—In maize-seedling fields, weeds and crop seedlings are small, visually similar, and often occluded amid drastic illumination and soil-background variation. We present *CDGL–YOLOv11Lite*, a lightweight YOLOv11-based detector that couples *Cross-Domain Shared Attention* (CDSA)—dual spatial/channel axial branches with shared cross-domain projections and post-alignment reweighting—with a *Conditional Gated Linear Unit* (CGLU) using depthwise 3×3 context-aware gating to suppress background misactivations and strengthen small-object cues. To satisfy edge constraints, we perform LAMP-based joint channel/layer pruning with light fine-tuning, then export to ONNX and optimize in TensorRT via operator fusion and mixed-precision inference. On a custom maize-weed dataset, *CDGL–YOLOv11Lite* achieves 90.25 % AP, 88.37 % recall, and mAP@0.5 of 90.45 %; the deployable model is 1.9 MB (0.8 M parameters, 3.1 GFLOPs) and runs at 67 FPS on NVIDIA Jetson Xavier NX, offering a practical balance of accuracy, efficiency, and edge-side deployability.

Index Terms—Maize-seedling weed detection, cross-domain shared attention, conditional gated linear unit, model pruning, TensorRT acceleration

I. INTRODUCTION

Rapid and accurate identification of seedling-stage weeds underpins variable-rate spraying, autonomous unmanned machinery, and digital farmland management. Unlike controlled indoor settings, field conditions exhibit pronounced domain shift (soil properties, moisture, ground cover, imaging devices), non-uniform illumination (backlighting, highlights, shadows), dynamic perturbations (wind/motion blur), and frequent occlusions. Maize seedlings and multiple weed species are morphologically and chromatically similar, fine-grained, and small-scale, which drives higher false positives/negatives and poorer cross-domain generalization for traditional methods and generic vision models [1]. Under tight on-device compute, power, and memory budgets along the smart-agriculture pipeline, detectors must jointly optimize accuracy, latency, and deployability.

Single-stage detectors (YOLO family) provide a strong speed-accuracy trade-off and mature tooling for implemented, mobile, and edge devices [2]; cross-scale fusion

(FPN/BiFPN) improves small-object separability and robustness in complex backgrounds [3], [4]; lightweight channel and channel-spatial attention (e.g., SE, ECA) enhances discriminative features with minimal overhead [5], while axial-decomposition attention retains long-range dependency modeling near linear complexity [6]; gated linear units (and convolutional variants) increase pathway selectivity and stability for dense small objects [7]; and structured pruning (channel/layer) reduces parameters and FLOPs with minor accuracy loss [8], which, paired with edge inference engines, enables real-time deployment [9]. Building on these insights and the data realities of long-tailed class distributions, inter-class similarity, and background co-occurrence under domain shift and illumination noise [10], we propose a lightweight, real-time field detector for maize seedlings, **CDGL-YOLOv11Lite**. It integrates (i) *Cross-Domain Shared Attention (CDSA)*—dual spatial and channel axial attention in backbone/neck with shared query-key projections to align cross-domain statistics, delivering near-linear reweighting that strengthens weak-texture/boundary cues and suppresses background misactivations for improved small-object recall and cross-plot robustness; (ii) a *Conditional Gated Linear Unit* whose gating branch uses 3×3 depthwise-separable plus pointwise convolutions to condition gates on local context, suppress background triggers, and better match edge parallelism/memory access; and (iii) *LAMP-based structured pruning and deployment*, where staged LAMP pruning removes redundancy and, together with operator fusion, layer folding, and FP16 (optional INT) quantization in TensorRT, reduces latency and memory on Jetson-class devices for practical field deployment.

II. RELATED WORK

Field maize seedling-weed detection on spray booms, ground carts, and UAVs must run under tight latency/power on Jetson-class devices, sustaining > 30 fps despite platform speed (0.8–1.2 m/s) and field-of-view jitter. Edge-friendly backbones and attentions improve global dependency modeling and channel-spatial recalibration at modest cost: Focal Modulation avoids explicit self-attention while retaining long-range context [11], MobileViT injects transformer blocks into compact CNNs [12], MaxViT scales receptive fields via multi-axis attention [13], and Coordinate Attention encodes positional cues effective for elongated seedling structures [22].

On the detector side, the YOLO line advances the speed-accuracy frontier: YOLOX updates label assignment and decoupled heads [14], YOLOv7 integrates trainable freebies [15], and PP-YOLOE streamlines neck/head and training for higher throughput [16]; recent YOLOv8/9/10 iterations further refine architectures and objectives for real-time deployment [19]–[21]. In parallel, DETR variants are now practical on edge GPUs: RT-DETR enables end-to-end transformers in real time [17], and DINO stabilizes training with improved denoising/anchors [18]. Localization-quality-aware objectives boost recall under jitter and small-object clutter: VarifocalNet couples classification with predicted IoU [23], Generalized Focal Loss v2 refines quality estimation [24], and Alpha-IoU/SIoU

strengthen box-regression gradients [25], [26]. Together, these form a toolbox for high-recall, low-latency detection.

Agricultural perception adds class similarity, weak textures, occlusion, and strong illumination/soil variability. Community resources emphasize realism and embedded constraints: YOLOWeeds benchmarks multi-class cotton weeds on embedded settings [27], Weed-AI aggregates expert-labeled diverse scenes [28], and rice disease corpora (e.g., Paddy Doctor) support field evaluation [29]. Plant-oriented ViTs (PlantXViT) are emerging [30], and surveys synthesize advances and gaps in plant pathology [31]. To bridge plot/device/time shifts without source data, source-free adaptation for detection is gaining traction, including YOLO-style learners [32]–[34]. Robustness-friendly training adds little overhead: Simple Copy-Paste improves multi-instance scenes [35], TrivialAugment offers competitive gains without search [36], and label-efficient learning is key under costly, seasonal annotation [37].

Guided by these insights, **CDGL-YOLOv11Lite** integrates mobile-friendly global dependency modeling and lightweight channel-spatial recalibration with real-time YOLO-style detection and quality-aware objectives, complemented by pragmatic augmentation/adaptation. Structured training and engine-level optimizations sustain high recall on embedded hardware for field maize seedling-weed scenarios.

III. MATERIALS AND METHODS

A. Data Sources and Processing

We collected data under a “multi-plot, multi-period, cross-device” protocol to reflect real field conditions, spanning soil textures (sandy loam, loam, clay), surface cover (bare soil, straw residue, withered weeds), moisture states (dry, moist, post-rain), and diverse lighting (direct sun, overcast, dusk backlight, fast cloud shadows). UAV/handheld and mobile industrial cameras were used with height/pitch adjusted to capture inter- and intra-row details. For deployability, detection was simplified to two classes—*maize* and *weed*. The dataset comprises 3,500 YOLO-format images, double-blind annotated and cross-checked by two agronomy-trained annotators; a “minimum discernible boundary” rule avoided merging neighbors or truncating edges. Stratified sampling by plot/time/device produced train/val/test splits of 7:2:1 (2,450/700/350) with domain orthogonality. Images were resized to 640×640 via letterbox, and effective-region masks were recorded so the loss accounts for boundary effects. To enhance robustness to reflections, hard shadows, mud splashes, straw textures, and wind motion, we combined standard color (HSV/brightness/contrast) and geometric transforms (scaling/translation/mild affine) with synthesized structured occlusions and strip-like shadows; weak Mosaic, limited MixUp, and same-class Copy-Paste (scale-constrained for small objects) were applied [38]. A hard-example sampler upweighted small/low-contrast *weed* instances, and a class-alignment term curbed threshold drift under batch fluctuations. This pipeline improved cross-plot/time/device generalization without increasing inference latency. Representative samples

exhibiting these diverse field conditions and annotations are visualized in Figure 1.

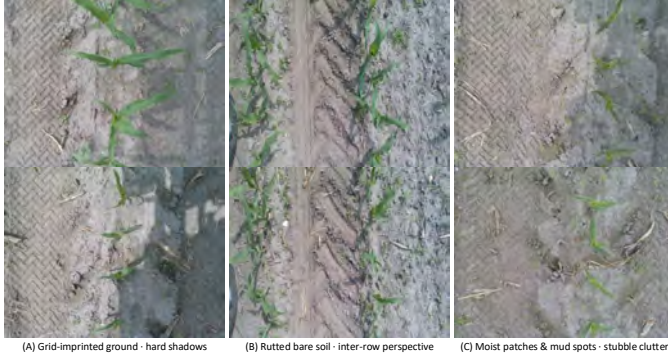


Fig. 1. Examples from the field maize-weed dataset.

B. Model Improvement

YOLO (*You Only Look Once*), introduced by Redmon *et al.* in 2016, formulates detection as end-to-end regression and delivers high throughput versus proposal-based two-stage methods; through successive iterations, the family has improved engineering and accuracy, with YOLOv11 refining architecture, feature fusion, and training (e.g., more rational C3k2 cross-stage aggregation, enhanced multi-scale fusion), alongside better losses and adaptive sample assignment to raise mAP while retaining real-time performance. However, under complex field conditions (strong reflections, hard shadows, mud splashes, straw residues, wind-induced motion) and with maize-weed targets that are small, similar, and often occluded, conventional feature extraction/fusion is prone to false positives and misses: mid/low-level features lack long-range dependency and domain-shift awareness for cross-plot/season/device generalization, and insufficient local-context selectivity triggers soil/residue misactivations. To address this, we propose **CDGL-YOLOv11Lite**, an improved YOLOv11 variant that replaces C3k2 repeat units with a lightweight composite of *Cross-Domain Shared Attention* (CDSA) and *Conditional Gated Linear Unit* (CGLU); concretely, C3k2 units in backbone P3/P4 aggregation blocks and the first bottom-up neck stage are substituted by CDSA-CGLU, jointly achieving global alignment recalibration and local context gating at minimal structural cost (Fig. 2).

C. C3k2_CDSA_CGLU Module

To improve small-object recall and cross-domain robustness without changing the outer topology or scale decoding, we replace all C3k2 modules with **C3k2_CDSA_CGLU** (CDGL) blocks. CDGL keeps the C3k2 “branch-aggregation-projection” pattern: after a lightweight stem (conv+norm), features split into a main branch that stacks n residual units (CDSA followed by CGLU) and an identity shortcut; the two are concatenated and projected by a 1×1 conv to restore channels, remaining interface-compatible with C3k2. For drop-in reuse across

backbone and neck, CDGL preserves input/output channels, stride, and residual topology, yielding consistent statistical alignment and selective suppression across scales. **CDSA** adopts axial decomposition with dual spatial/channel branches, computing windowed row/column correlations and fusing them to capture long-range dependencies at near-linear cost in mid/low-level features; to mitigate plot/season/device drift, its query-key projections are shared or low-rank coupled across same-scale layers and lightly affine-calibrated for domain-aligned global recalibration [39]. **CGLU** uses a DWConv $3 \times 3 + \text{PW } 1 \times 1$ gated branch to derive position-wise gates from local context, suppressing false activations from straw textures, mud spots, and hard shadows while preserving fine edges and weak-texture leaf details; units are cascaded as “CDSA $\rightarrow$ CGLU,” each with a short residual, and a final 1×1 projection provides channel reduction/feedback to ensure strict C3k2 compatibility. In terms of complexity and deployment, CDGL relies on standard convolutions and per-channel ops: CDSA’s axial correlation runs in $\mathcal{O}(L)$ with window length L , and CGLU’s main overhead (pointwise conv) is fusable at inference. Consequently, even after full replacement, the overall budget is nearly unchanged, while unified projection/gating across scales harmonizes feature selectivity and statistics along multi-scale, bidirectional paths, reducing cross-domain degradation and background false alarms (Fig. 3).

D. Cross-Domain Shared Attention Mechanism (CDSA)

To handle weak textures, small scales, and noisy backgrounds (soil, straw, mud) in field maize-weed scenes, we adopt a **Cross-Domain Shared Attention (CDSA)** module (Fig. 4). CDSA axially decomposes spatial and channel attention to balance global statistical alignment with local detail enhancement at near-linear cost.

Let $f \in \mathbb{R}^{C \times H \times W}$ be the input. A shared base and branch split are:

$$F_{\text{share}} = \text{ReLU}\left(C_{3 \times 3}(\text{ReLU}(C_{1 \times 1}(f)))\right), \quad (1)$$

$$(F_1, F_2) = \text{Split}(f). \quad (2)$$

Spatial branch. Horizontal/vertical context is aggregated via channel-wise max/avg pooling; a 1×1 conv and sigmoid yield mask M_s and refined feature P_s :

$$M_s = \sigma\left(C_{1 \times 1}(\text{max}((F_{\text{share}}), (F_{\text{share}})))\right), \quad (3)$$

$$P_s = F_1 \odot M_s. \quad (4)$$

Channel branch. Global average pooling gives $z = (F_2)$. A kernel set $\mathcal{K} = \{k_i\}_{i=1}^n$ adapts interaction range:

$$k_1 = \left\lfloor \frac{\log_2 C}{\gamma} + b \right\rfloor_{\text{odd}}, \quad k_i = k_1 + 2(i-1), \quad i = 2, \dots, n. \quad (5)$$

Channel weights and refinement:

$$w = \sigma\left(\sum_{k \in \mathcal{K}} k(z)\right), \quad P_c = F_2 \odot w. \quad (6)$$

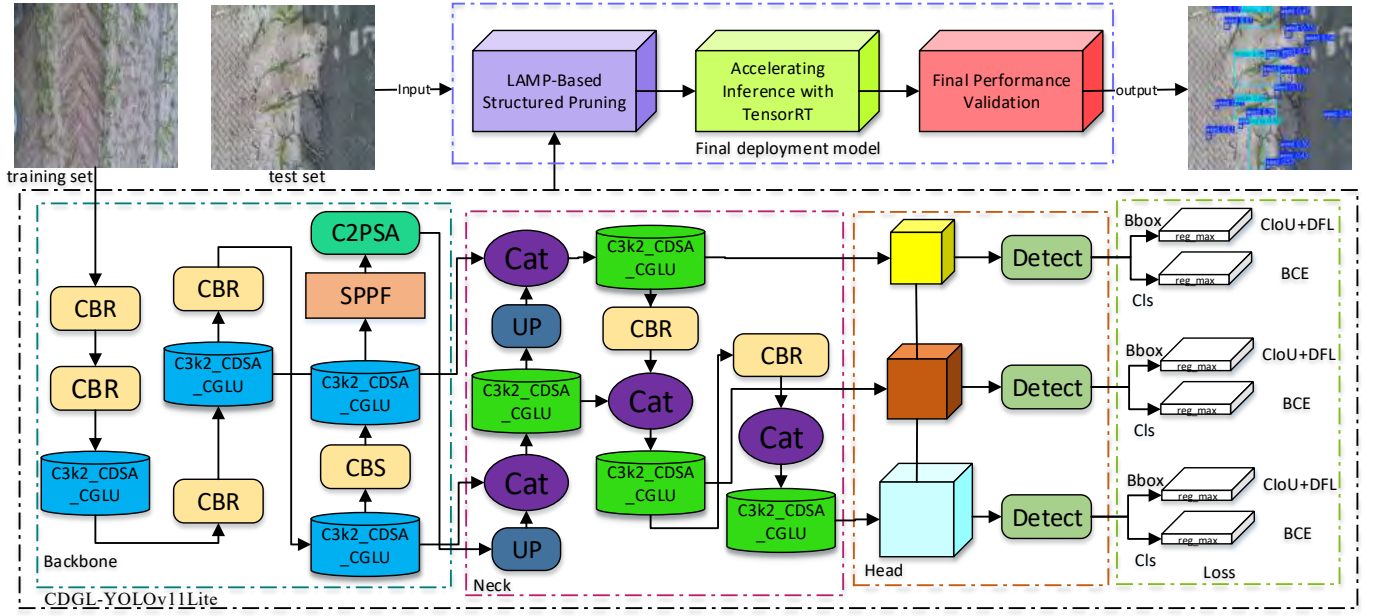


Fig. 2. Overall architecture of the proposed CDGL-YOLOv11Lite model.

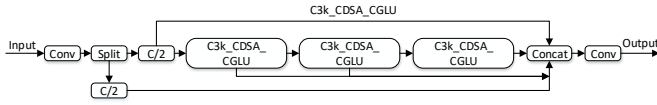


Fig. 3. Structure of the C3k2_CDSA_CGLU (CDGL) block.

Fusion. Spatially and channel-enhanced features are fused and residually recalibrated:

$$F_a = \text{LRReLU}\left(C_{3 \times 3}((P_s, P_c))\right), \quad F_{\text{CDSA}} = f \oplus F_a. \quad (7)$$

For efficiency, fusion uses only 1×1 and 3×3 kernels; GAP and affine scaling remain differentiable and TensorRT-friendly. CDSA improves robustness of YOLO-like detectors under complex illumination, cluttered backgrounds, and cross-device deployment while keeping overhead suitable for Jetson-class systems.

E. LAMP-based Channel Pruning

For high-speed, low-latency maize seedling-weed detection on edge devices, we reduce computation while preserving small-target performance via **LAMP** (*Layer-Adaptive Magnitude-based Pruning*). Unlike per-layer magnitude pruning (MP), whose statistics vary across scales and yield unbalanced compression, LAMP normalizes importance across layers to realize globally balanced structured sparsity.

Let convolutional weights be $W = \{W^{(l)}\}_{l=1}^L$. Define the importance for channel u as

$$\text{score}(u; W) = \frac{(W[u])^2}{\sum_{v \in W} (W[v])^2}. \quad (8)$$

Here the numerator measures the channel magnitude and the denominator normalizes by layer energy. To ensure cross-layer

fairness (avoiding bias toward small layers), LAMP enforces an order-preserving normalization:

$$(W[u])^2 \geq (W[v])^2 \Rightarrow \text{score}(u; W) \geq \text{score}(v; W). \quad (9)$$

Global pruning then ranks channels by score and removes them in batches to a target sparsity, automatically adapting to depth and energy without heuristic thresholds. Integrated into **CDGL-YOLOv11Lite**, LAMP prunes redundant channels across backbone, neck, and head while preserving critical pathways for small objects. Post-pruning, brief fine-tuning with distillation restores balance, and TensorRT layer fusion further cuts parameters and latency. Overall, we retain accuracy/robustness with $> 50\%$ parameter reduction and $\sim 1.8\times$ inference acceleration on Jetson-class devices.

F. TensorRT Deployment Optimization

TensorRT converts trained graphs into hardware-optimized engines via operator fusion and constant folding (e.g., conv+norm+activation), reducing memory traffic; at the graph level it performs layout planning, kernel auto-tuning, and unified memory management. Mixed precision (FP16) increases throughput; INT8 uses calibrated ranges for stability. Bounding-box decoding, confidence filtering, and NMS are embedded as plugins so inference and post-processing remain on-GPU, avoiding host-device transfers.

For deployability, **CDGL-YOLOv11Lite** follows *fusion-/quantization-friendly* and *shape-consistent* design: CDSA/CGLU use only $1 \times 1/3 \times 3$ convs, GAP, and elementwise ops (TensorRT-friendly). After pruning, main and shortcut branches are synchronously trimmed by identical channel indices; multi-scale up/down paths keep consistent channel counts to prevent concat mismatches. The detection head exports as a single block connected to the NMS plugin.

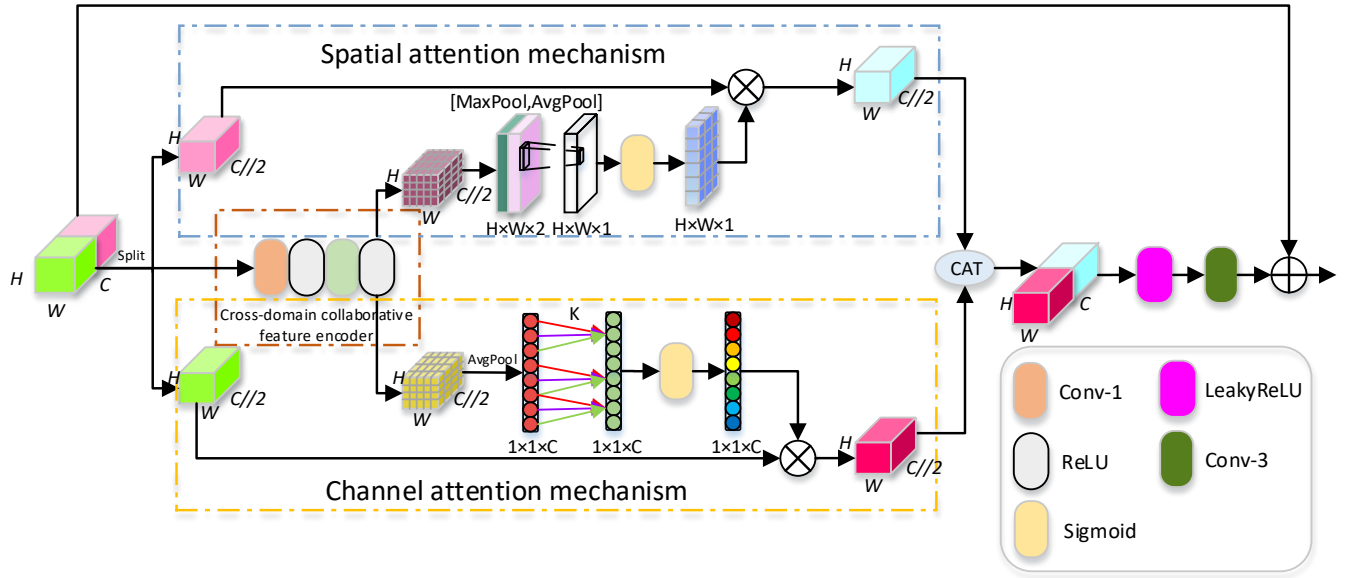


Fig. 4. Architecture of the Cross-Domain Shared Attention (CDSA) module.

During engine building, fixing input and batch sizes speeds kernel search; treat depthwise and pointwise convs with distinct search constraints. For INT8, calibrate on representative field samples (strong reflections, shadows, occlusions) to stabilize gating/attention numerics. These strategies enable a compact, unified engine on Jetson Xavier NX, delivering high-speed, low-latency, energy-efficient weed detection.

IV. MODEL TRAINING AND RESULT ANALYSIS

A. Experimental Environment Configuration

To support continuous field monitoring and evaluation, we built an interactive data-collection/annotation platform and trained the initial models on a high-performance workstation. Hardware/software: NVIDIA GeForce RTX 5090 (GPU), Ubuntu 18.04 LTS, PyTorch 2.1.0, PyCharm 2025.3, OpenCV 4.4.6, Python 3.7. For edge deployment, models were exported to ONNX and compiled with TensorRT (FP16) on an NVIDIA Jetson Xavier NX. Decoding and post-processing were embedded as plugins to minimize host-device transfers. An interactive UI displays real-time results and logs field data (Fig. 5).

Training used SGD (momentum 0.937, weight decay 0.0005). The learning rate followed a two-stage schedule: 0.01 for the first 200 epochs, decaying to 0.0001 over the subsequent 150 epochs; a 3-epoch linear warm-up stabilized early optimization. This configuration balanced convergence efficiency with deployability and interactive usability on edge devices.

B. Experimental Results and Analysis

We compared YOLOv8n, YOLOv13n, YOLOv11n, and the proposed **CDGL-YOLOv11n**.

As shown in Fig. 6 and Table I, **CDGL-YOLOv11n** achieves the best overall performance (AP 90.25%,



Fig. 5. User interface of the CDGL-YOLOv11Lite detection system.

TABLE I
PERFORMANCE COMPARISON ON MAIZE-WEED DETECTION.

Model	AP (%)	AR (%)	mAP@0.5 (%)	Size (MB)	Time (s)
YOLOv8n	87.56	87.13	87.59	6.2	0.023
YOLOv13n	88.01	86.11	87.35	5.2	0.044
YOLOv11n	88.13	86.78	87.98	5.3	0.022
CDGL-YOLOv11n	90.25	88.37	90.45	5.5	0.022

AR 88.37%, mAP@0.5 90.45%) while keeping latency identical to YOLOv11n (0.022 s) and adding only ~0.2 MB. Gains are most evident for (1) weak-contrast, fine-edged small targets and (2) strong background co-occurrence (soil, straw, hard shadows), benefiting from CDSA's spatial-channel recalibration and CGLU's suppression of background misactivations. In contrast, YOLOv8n is larger without superior accuracy, and YOLOv13n is slower without accuracy gains. Overall, **CDGL-YOLOv11n** offers the most balanced real-time option for edge-side agricultural deployment.

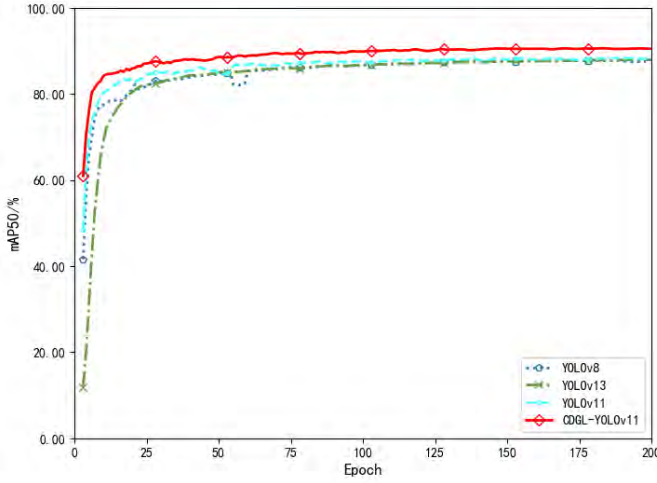


Fig. 6. mAP@0.5 comparison on the field maize-weed test set.

TABLE II
FEATURE-LEVEL DOMAIN SHIFT (MEAN $\pm$ STD, $R=5$). LOWER IS BETTER
FOR MMD/CORAL/ $\hat{A}$.

Layer	MMD $\downarrow$	CORAL $\downarrow$	$\hat{A}\downarrow$
YOLOv11n-P3	0.112 $\pm$ 0.008	0.286 $\pm$ 0.019	1.42 $\pm$ 0.06
CDSA-P3	0.087$\pm$0.007	0.209$\pm$0.015	1.11$\pm$0.08
YOLOv11n-P4	0.096 $\pm$ 0.007	0.241 $\pm$ 0.017	1.35 $\pm$ 0.07
CDSA-P4	0.074$\pm$0.006	0.192$\pm$0.014	1.05$\pm$0.07
YOLOv11n-P5	0.082 $\pm$ 0.006	0.217 $\pm$ 0.015	1.28 $\pm$ 0.05
CDSA-P5	0.069$\pm$0.005	0.183$\pm$0.012	1.01$\pm$0.06

C. Cross-Domain Generalization and Gating Insights

Setup. To reflect real deployment, domains are defined by plot (A/B/C), season (T1/T2), and device (D1/D2), with (A,T1,D1) as *source* and (BUC,T2,D2) as *target*; splits are image-disjoint and domain-orthogonal. We quantify feature shift at pyramid levels P3/P4/P5 using MMD (RBF), CORAL (covariance Frobenius), and an A-distance proxy $\hat{A}=2(1-2\epsilon)$ from a domain-classifier error ϵ (lower is better). Detection is evaluated by AP@0.5, mAP@0.5:95, small-object AP_S, and background FPPI (lower is better). Protocols include: (i) S $\rightarrow$ T zero-shot; (ii) S $\rightarrow$ T with 5% labeled target fine-tuning; and (iii) S $\rightarrow$ T source-free adaptation (SFDA) via confidence-thresholded self-training with an EMA teacher, using identical schedules across methods [32]–[34]. The resulting feature distribution discrepancies between the source and target domains are quantified in Table II.

Findings. At the feature level, CDSA consistently narrows source–target gaps across P3/P4/P5 (e.g., P3 MMD 0.112 $\rightarrow$ 0.087, CORAL 0.286 $\rightarrow$ 0.209), indicating improved alignment that is most beneficial where small-object cues are formed. This translates into better cross-domain detection: CDGL improves AP@0.5, mAP_{0.5:0.95}, AP_S, and FPPI under zero-shot, 5% fine-tuning, and SFDA protocols without extra latency (Table III). In parallel, lightweight gating design choices favor edge deployment: compared with recent modules (SimAM [10], Coordinate Attention [22], Triplet

TABLE III
CROSS-DOMAIN DETECTION (S $\rightarrow$ T). AP IN %; FPPI LOWER IS BETTER.

Method	AP@0.5	mAP _{0.5:0.95}	AP _S	FPPI $\downarrow$
YOLOv11n (zero-shot)	83.1	48.6	61.4	0.52
CDGL-YOLOv11n (zero-shot)	86.3	51.2	66.8	0.36
YOLOv11n (+5% fine-tune)	85.7	50.8	65.1	0.44
CDGL-YOLOv11n (+5% FT)	88.6	53.9	70.5	0.31
YOLOv11n (SFDA)	84.9	50.1	64.1	0.41
CDGL-YOLOv11n (SFDA)	87.8	53.0	69.1	0.33

TABLE IV
CDGL VARIANTS UNDER PRUNING AND TENSORRT OPTIMIZATION.
PARAMS/GFLOPS ARE PRE-ENGINE; FPS ON JETSON XAVIER NX
(FP16).

Model	Params [M]	GFLOPs	mAP _{0.5} (%)	FPS
YOLOv11n baseline	2.5	6.3	87.98	45 $\dagger$
CDGL-YOLOv11n	2.7	6.3	90.45	45 $\dagger$
+ LAMP pruning + fine-tune	0.8	3.1	89.84	50 $\dagger$
+ TensorRT (Lite)	0.8	3.1	88.74	67

Attention [9]), our CGLU (DW-3 $\times$ 3 + PW-1 $\times$ 1) provides stronger small-object selectivity and lower background false alarms with comparable latency—aligning with the model’s TensorRT-friendly, deployability-first objectives [11], [13]. Collectively, these results support the use of CDSA for domain alignment and CGLU for robust local suppression in real field conditions.

D. Lightweight Pruning and TensorRT Deployment

We evaluate the combined effects of LAMP channel pruning and engine-level optimization on CDGL-YOLOv11 for field maize-weed detection. *Procedure.* Channels are ranked per layer and normalized by tail energy to obtain cross-layer comparable scores; a single global threshold performs one-shot pruning, followed by short fine-tuning to restore distributional balance. Channel alignment is preserved across multi-scale fusion and residual branches. *Deployment.* The pruned model is exported to ONNX and compiled with TensorRT (operator fusion, constant folding, integrated post-processing) using FP16 to shorten the end-to-end path and reduce memory use. All experiments share the same input resolution and dataset split, maintaining outer-topology integrity while optimizing accuracy, compactness, and edge deployability. Table IV summarizes the quantitative trade-offs between model size, accuracy, and inference speed resulting from this pipeline.

Summary. LAMP delivers adaptive, structure-friendly compression (2.7 M $\rightarrow$ 0.8 M; 6.3 GFLOPs $\rightarrow$ 3.1 GFLOPs) with minimal mAP drop, and TensorRT further boosts edge throughput to 67 FPS while preserving practical accuracy and deployability.

E. Detection Performance of Different Models

To verify field effectiveness, **CDGL-YOLOv11Lite** was deployed on the detection platform and evaluated on the maize-weed test set, with five representative scenes compared against YOLOv11n: dense inter-row weeds, ruts/grid-textured

soil, strong backlighting with hard shadows, and mixed slender seedlings with small weeds. Overall, **YOLOv11n** shows more false positives under strong textures/uneven illumination and more misses or duplicate boxes near seedlings. In contrast, **CDGL-YOLOv11Lite** reduces irrelevant detections, yields boxes that better follow object contours, preserves row continuity, and more stably detects small weeds adjacent to stems or partially occluded by shadows, with fewer false alarms on healthy seedlings. Scene-wise, it separates closely adjacent small targets, suppresses high-frequency texture responses on tire-marked soils, and strengthens weak-contrast cues under backlighting/shadows via spatial-channel recalibration—reducing misses. These qualitative gains match prior quantitative results: without increasing latency or model size, cross-domain shared attention and conditional gating raise sensitivity to fine, low-contrast weeds while suppressing background noise, making **CDGL-YOLOv11Lite** suitable for real-time field inspection and variable-rate spraying (Fig. 7).

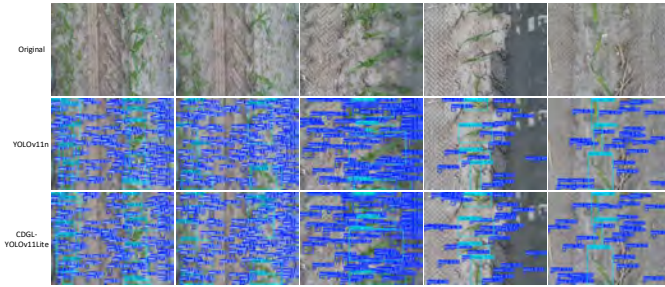


Fig. 7. Comparison of detection results between YOLOv11n and CDGL-YOLOv11Lite on field maize-weed scenes.

F. Heatmap Analysis of Detection Results

We interpret model attention via Grad-CAM++ [40] on final convolutional responses, comparing YOLOv11n (top) and **CDGL-YOLOv11Lite** (bottom) across dense inter-row weeds, ruts/grid-patterned soil, and strong backlighting with hard shadows (Fig. 8). **YOLOv11n** exhibits diffuse activations with broad high-response areas over soil cracks, tire marks, and textured grids, often yielding background false detections; saliency on small weeds around seedlings is weak and poorly localized. In contrast, **CDGL-YOLOv11Lite** produces compact hotspots concentrated on weed leaf tips and vein junctions, with responses forming continuous bands along crop rows; under strong reflections or shadows, background activations are markedly suppressed while small weeds near stem bases remain salient. These patterns indicate that **CDSA** strengthens spatial-channel recalibration and **CGLU** suppresses texture-induced noise, yielding higher sensitivity to weak-contrast, small, and densely clustered weeds and stronger restraint to complex backgrounds—favorable for real-time precision weeding and variable-rate spraying.

V. CONCLUSION

We propose **CDGL-YOLOv11Lite** for robust field maize-weed detection under complex conditions. The model

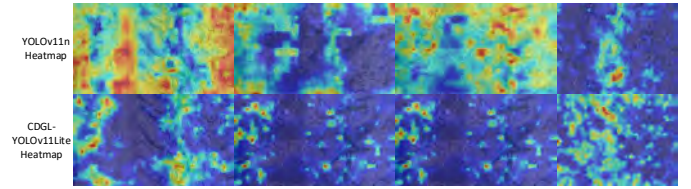


Fig. 8. Heatmap analysis of detection results for YOLOv11n and CDGL-YOLOv11Lite under various field conditions.

integrates **Cross-Domain Shared Attention (CDSA)** and **Conditional Gated Linear Unit (CGLU)** to enhance feature recalibration and suppress background noise. To ensure efficient edge deployment, we employ **LAMP** pruning and TensorRT optimization, which significantly reduce parameter count and latency while maintaining high accuracy. Experiments validate that the framework outperforms lightweight baselines, offering a balanced, real-time solution for precision agriculture on Jetson-class devices.

FOUNDING SOURCE

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